

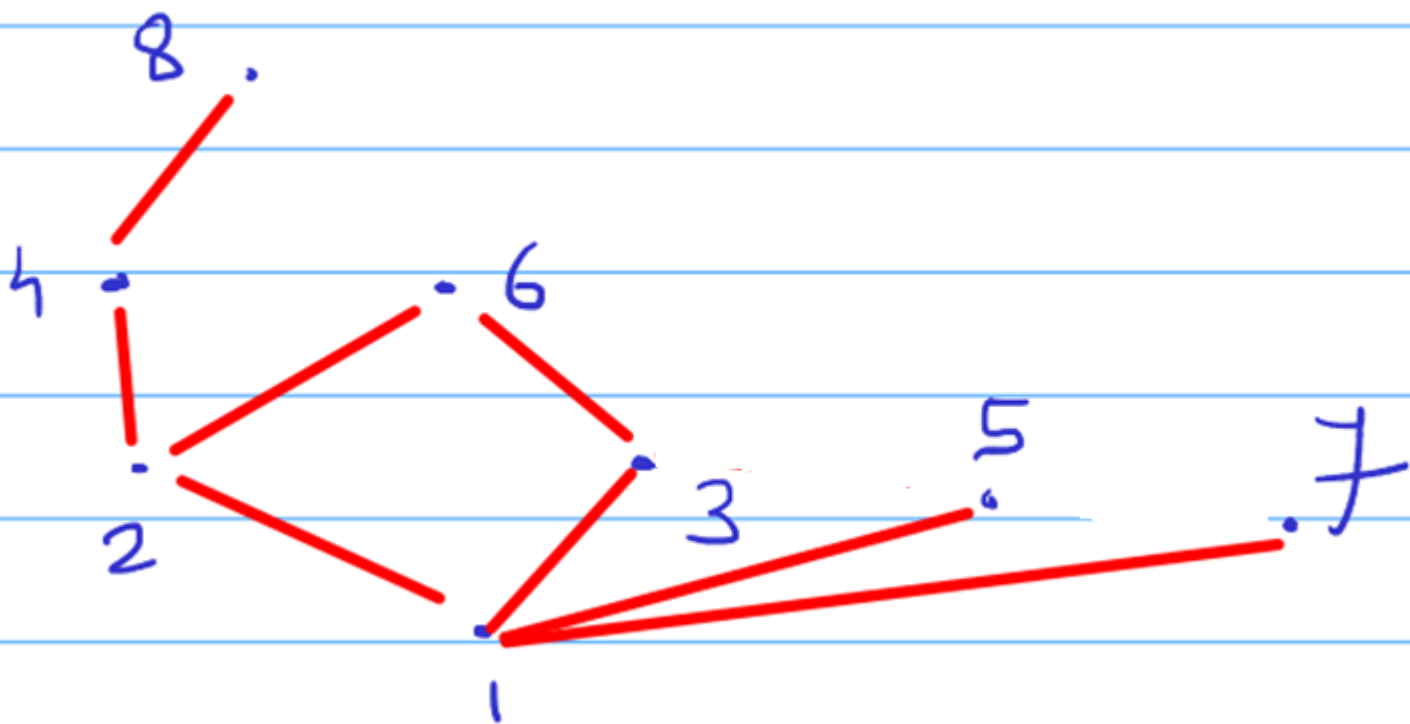
Hasse Diagram, Chain and Antichain

Maximal and Minimal Elements

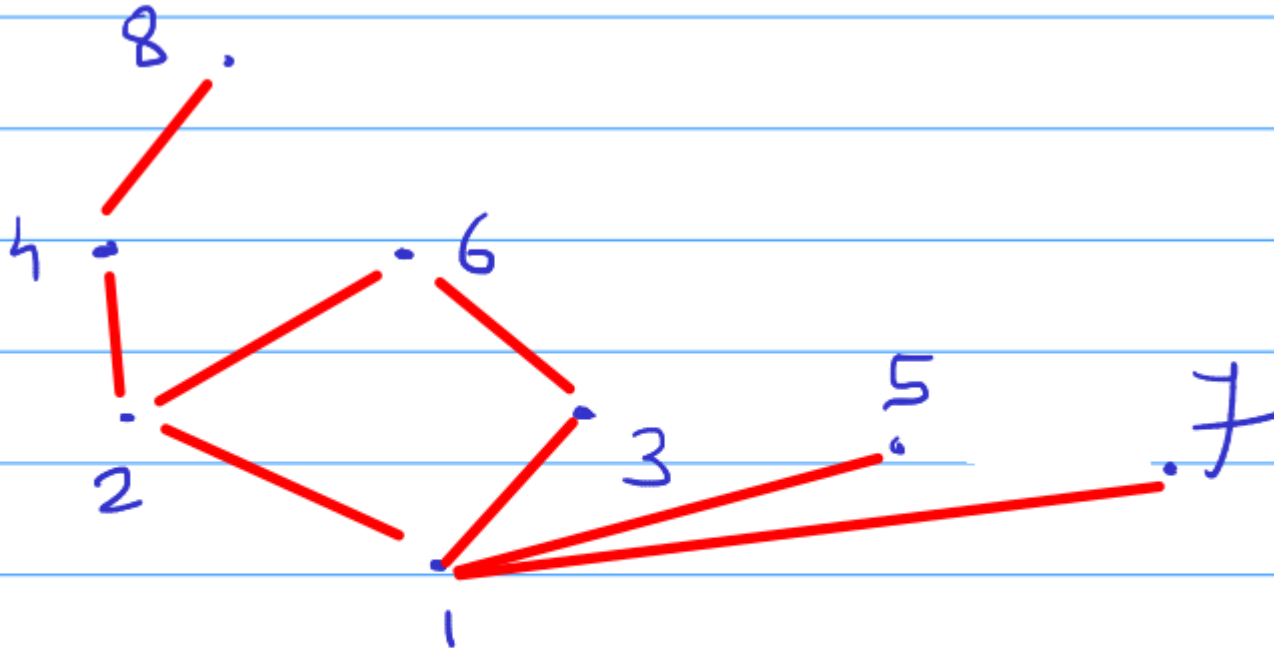
- **Maximal element** is an element of a POSET which is not less than any other element of the POSET. Or we can say that it is an element which is not related to any other element. Top elements of the Hasse Diagram.

Maximal and Minimal Elements

- **Minimal element** is an element of a POSET which is not greater than any other element of the POSET. Or we can say that no other element is related to this element. Bottom elements of the Hasse Diagram.



Maximal and Minimal Elements



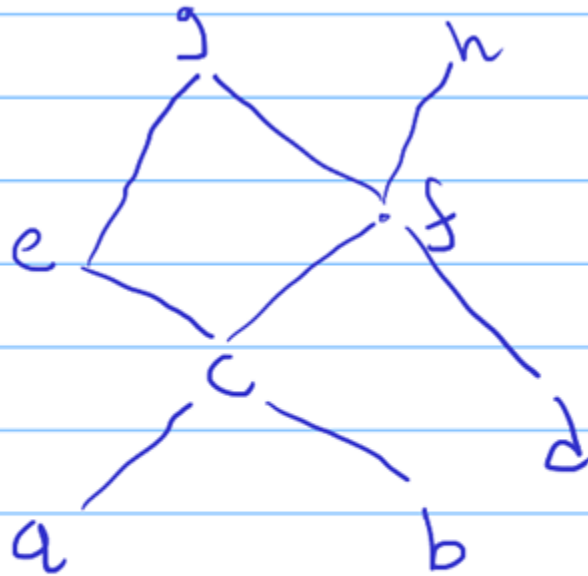
Maximal : 8, 6, 5, 7

Minimal : 1

Upper Bound and Lower Bound

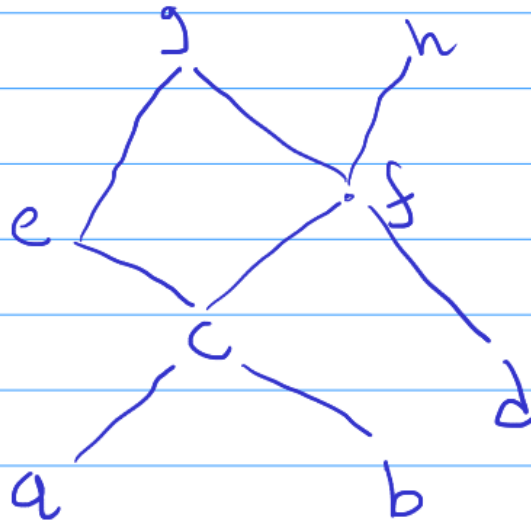
- **Upper Bound:** Consider B be a subset of a partially ordered set A . An element $x \in A$ is called an upper bound of B if $y \leq x$ for every $y \in B$.
- **Lower Bound:** Consider B be a subset of a partially ordered set A . An element $z \in A$ is called a lower bound of B if $z \leq x$ for every $x \in B$.

Example



$$B = \{e, c\}$$

Example

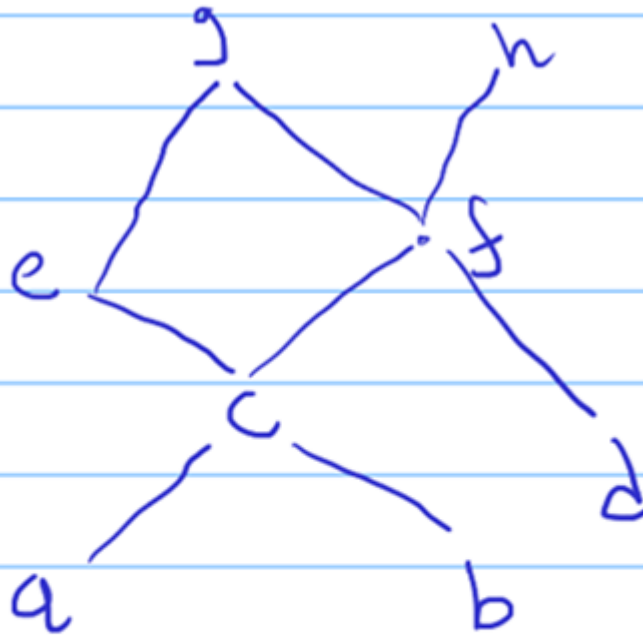


$$B = \{e, c\}$$

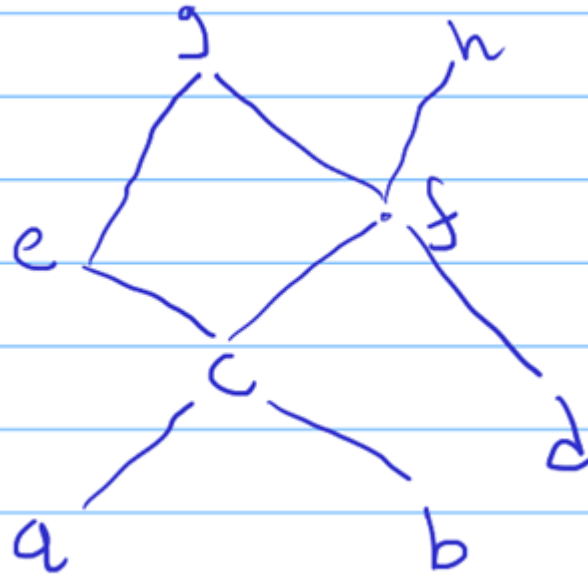
$$U.B(B) = \{g, e\}$$

$$L.B(B) = \{a, b, c\}$$

Example



$$B = \{c, f, d\}$$

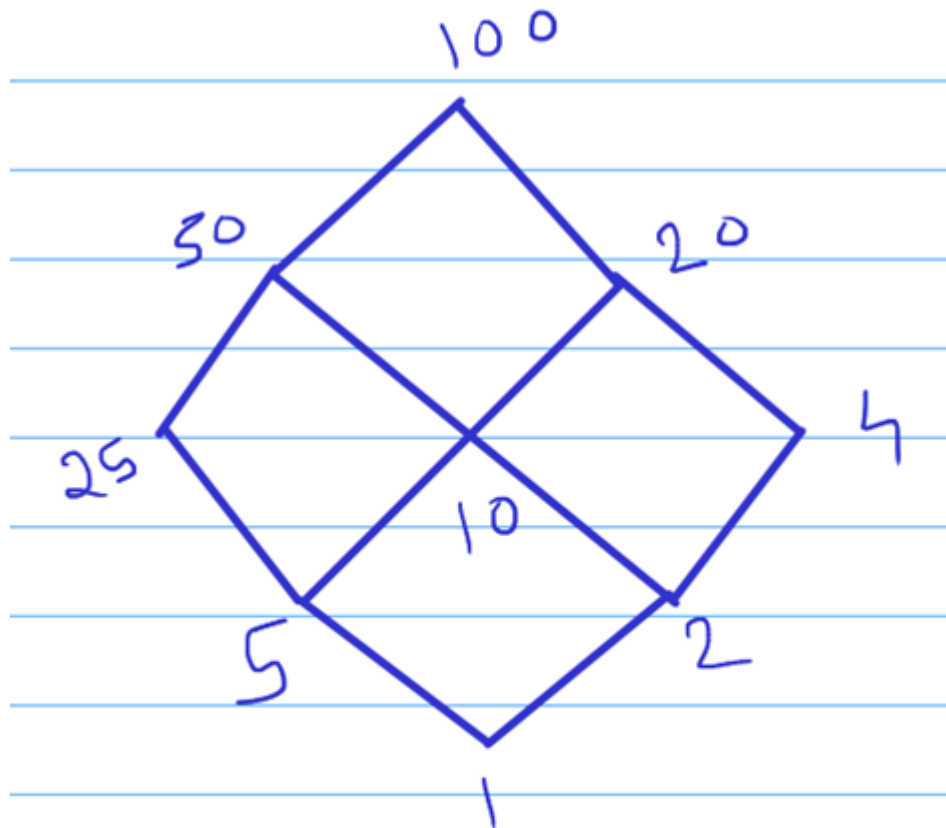


$$B = \{c, f, d\}$$

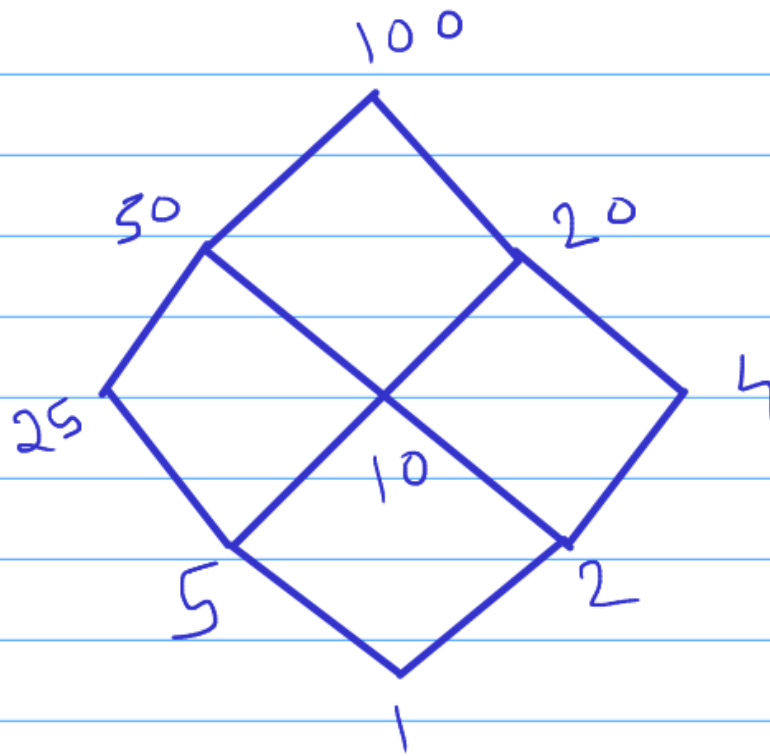
$$U.B(B) = \{g, h, f\}$$

$$L.B(B) = \emptyset$$

Example



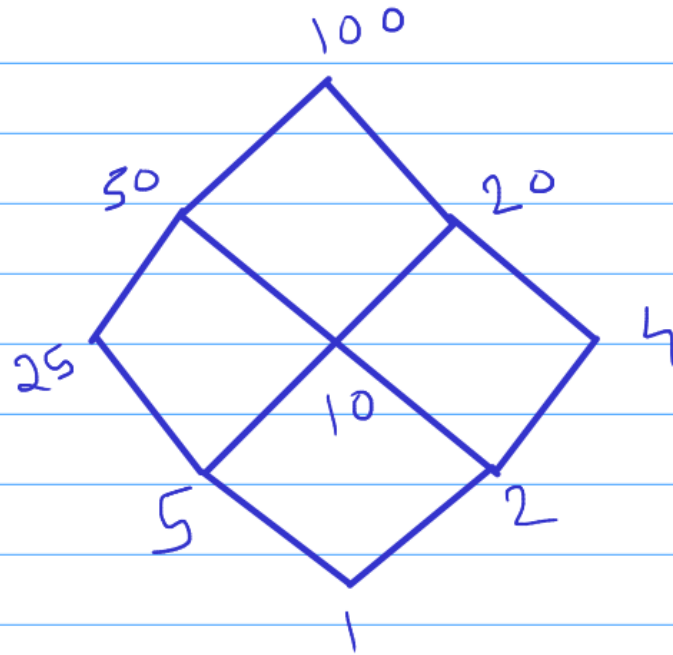
$$B = \{5, 10\}$$



$$B = \{5, 10\}$$

$$U.B(B) = \{50, 100, 20, 10\}$$

$$L.B(B) = \{1, 5\}$$

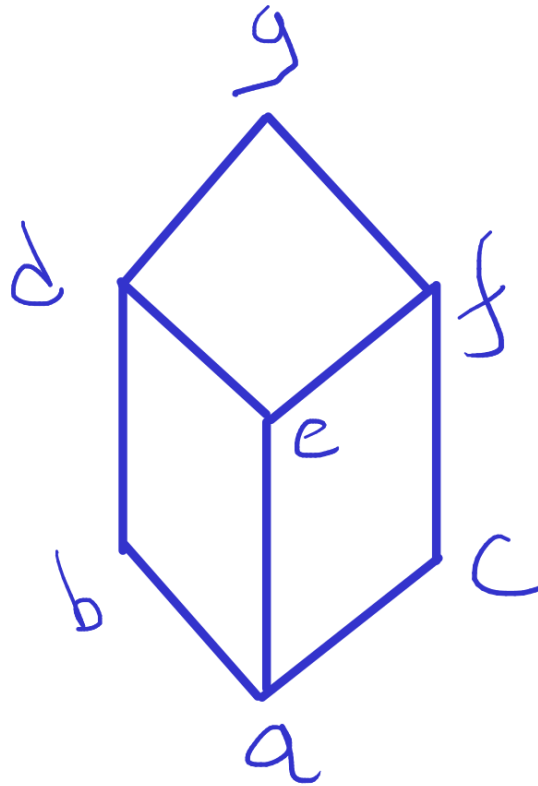


$$B = \{5, 10, 2, 4\}$$

$$B = \{5, 10, 2, 4\}$$

$$U.B(B) = \{20, 100\}$$

$$L.B(B) = \{1\}$$



$$\textcircled{1} B = \{d, g\}$$

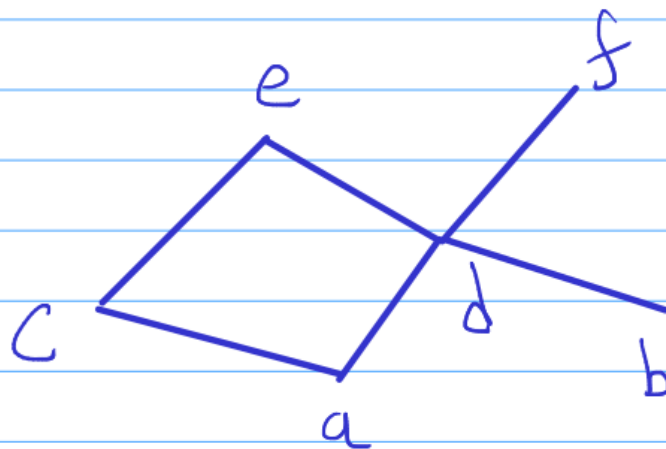
$$\textcircled{2} B = \{e, f\}$$

Least Upper Bound (LUB, Supreme, Join)

- Let A be a subset of a partially ordered set S . An element M in S is called an upper bound of A if M succeeds every element of A , i.e. if, for every x in A , we have $x \leq M$.
- **Least (Minimum) element in Upper Bound.**

Greatest Lower Bound (GLB, Infimum, Meet)

- An element m in a poset S is called a lower bound of a subset A of S if m precedes every element of A , i.e. if, for every y in A , we have $m \leq y$.
- **Greatest (Maximum) element in lower bound.**



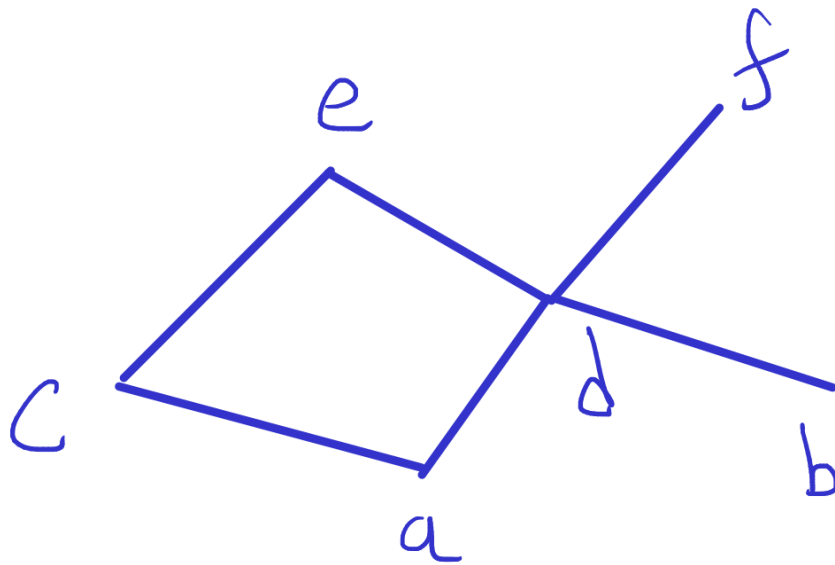
$$\textcircled{2} \quad B = \{c, d\}$$

$$U.B(B) = e$$

$$LUB(B) = e$$

$$L.B(B) = a$$

$$GLB(B) = a$$



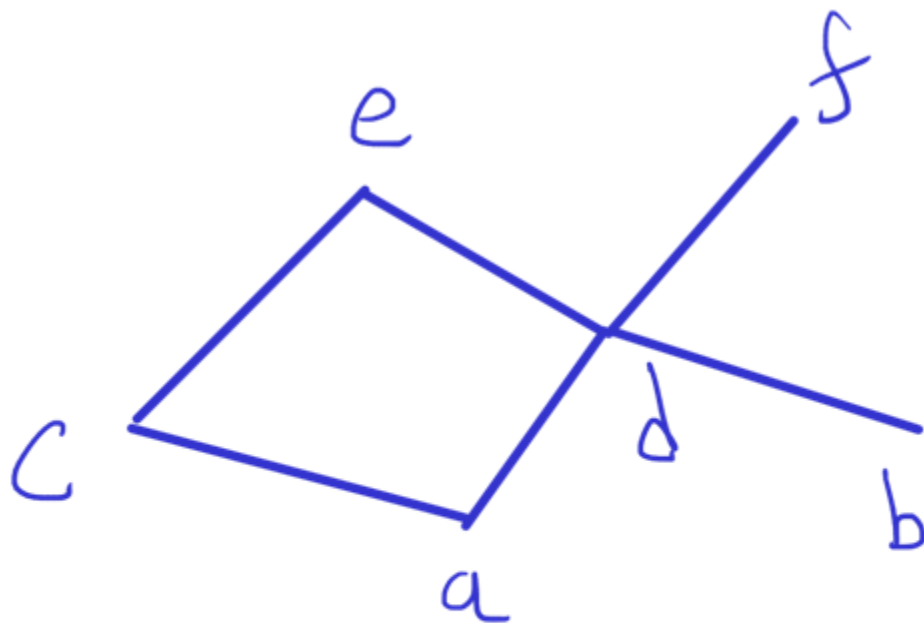
$$\textcircled{2} \quad B = \{a, b\}$$

$$U.B(B) = \{e, f, d\}$$

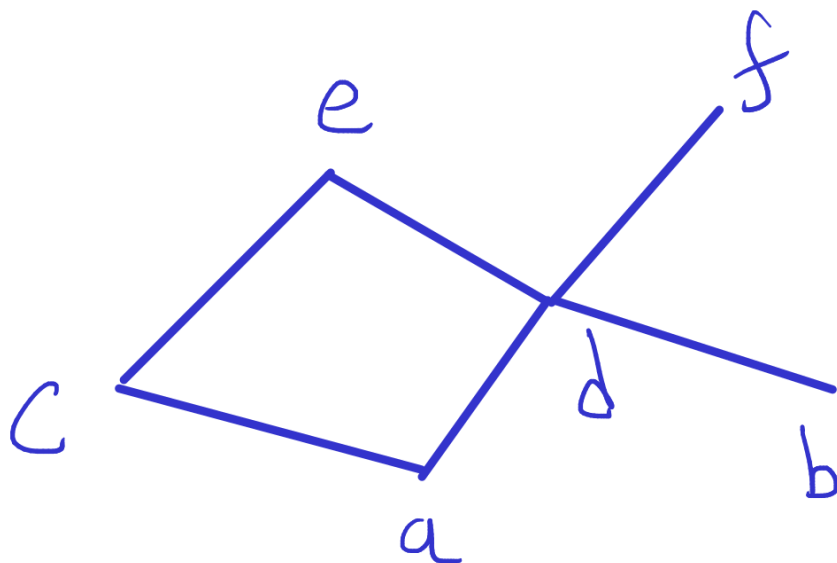
$$LUB(B) = d$$

$$L.B(B) = \emptyset$$

$$GLB(B) = \emptyset$$



③ $B = \{e, f\}$



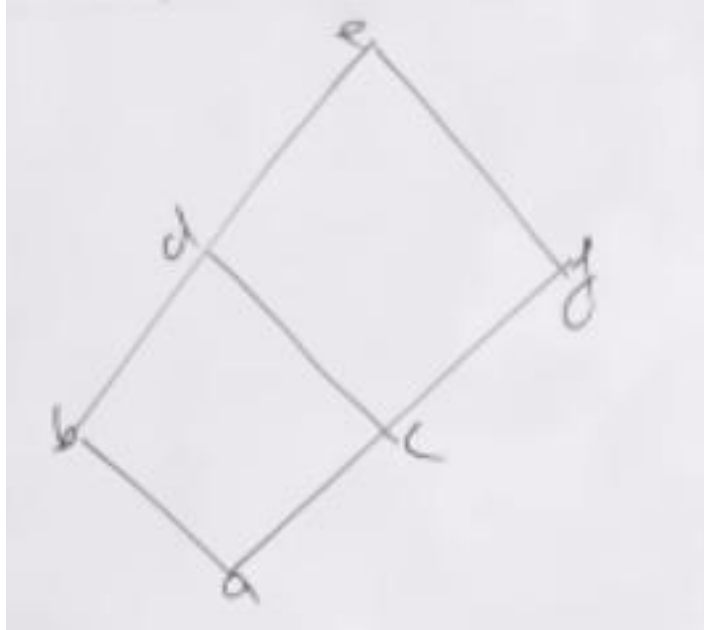
$$\textcircled{3} \quad B = \{e, f\}$$

$$U.B(B) = \emptyset$$

$$LUB(B) = \emptyset$$

$$LB(B) = \{a, d, b\}$$

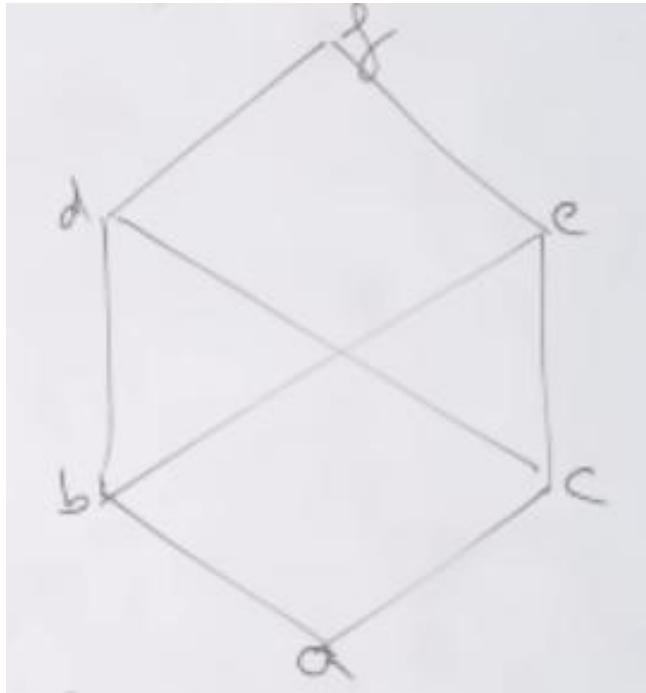
$$GLB(B) = d$$



1. $B = \{ a, c, f \}$

2. $B = \{ d, c \}$

Find LUB and GLB.



1. $B = \{ d, e \}$

2. $B = \{ b, c \}$

Find LUB and GLB.

Lattice

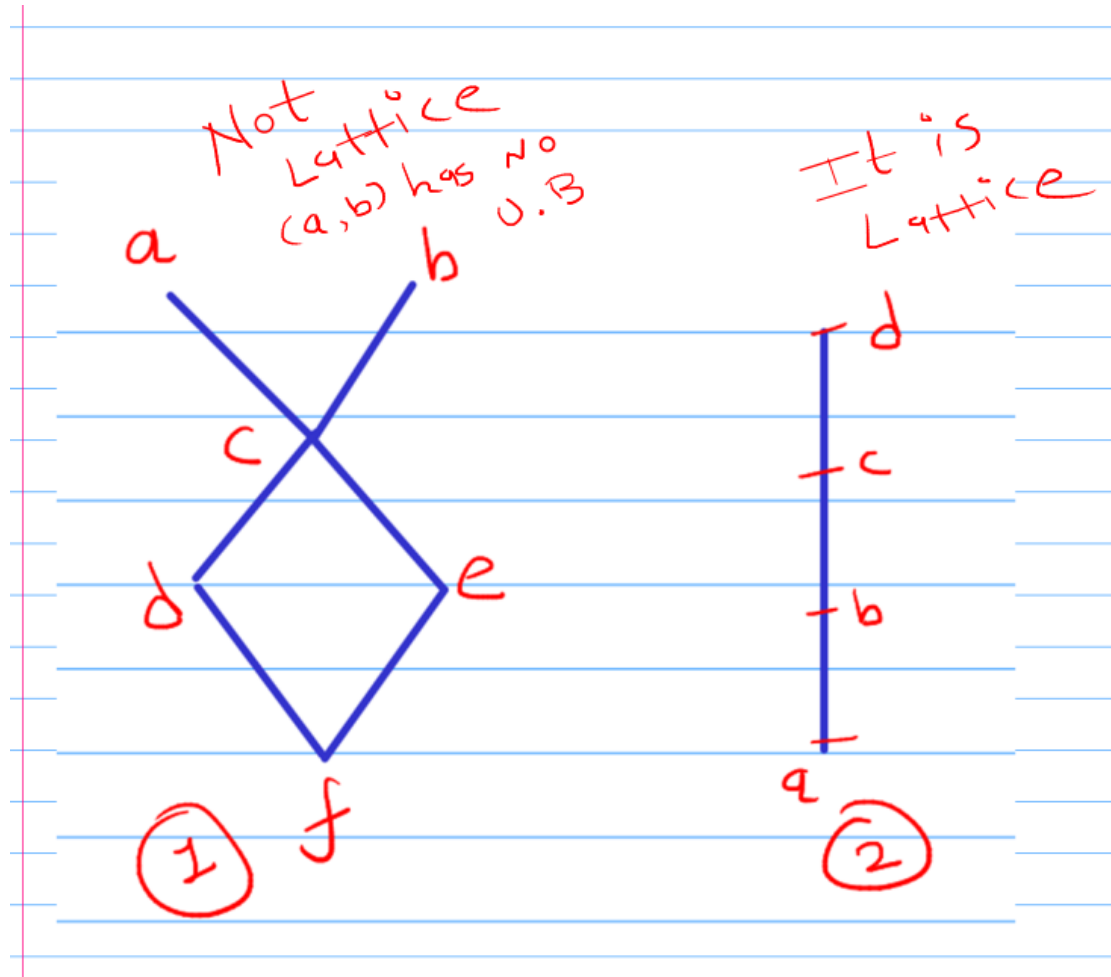
- A partially ordered set is said to be a lattice if every two elements in the set have a least upper bound and a greatest lower bound.

i.e.,

$\forall x, y \in S, \text{GLB}(x, y) \neq \emptyset$ and

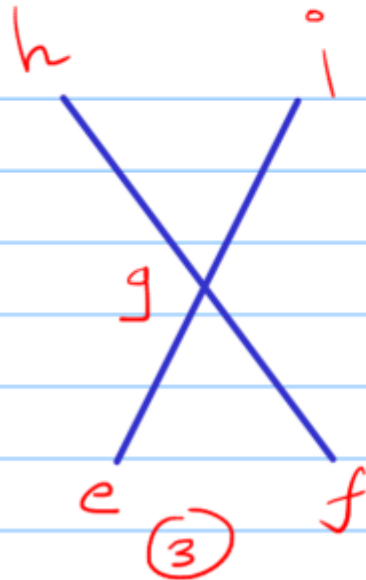
$\forall x, y \in S, \text{LUB}(x, y) \neq \emptyset$

Identify whether below structures are lattice or not ?

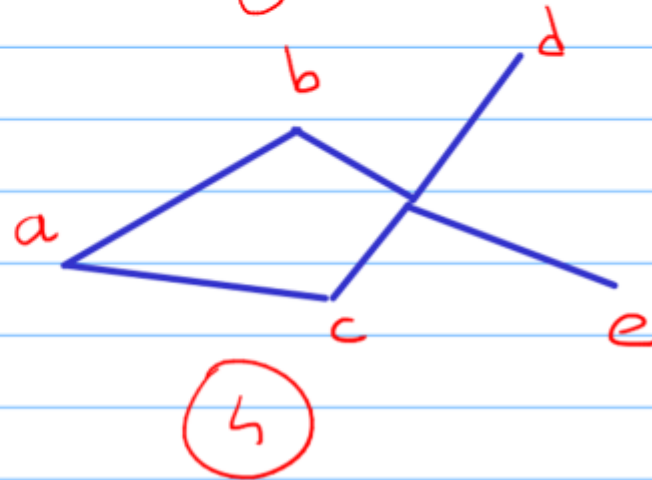


Identify whether below structures are lattice or not ?

Not Lattice
 $U.B(h,i) = \phi$

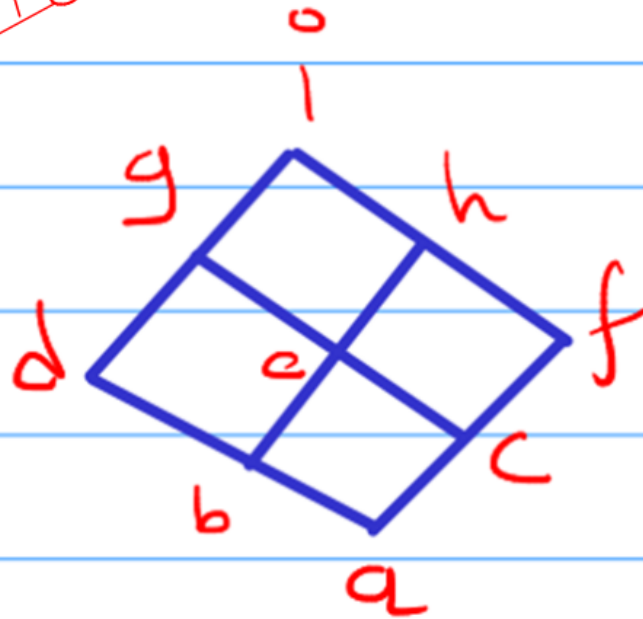


Not Lattice
 $U.B(b,d) = \phi$



Identify whether below structures are lattice or not ?

It is Lattice



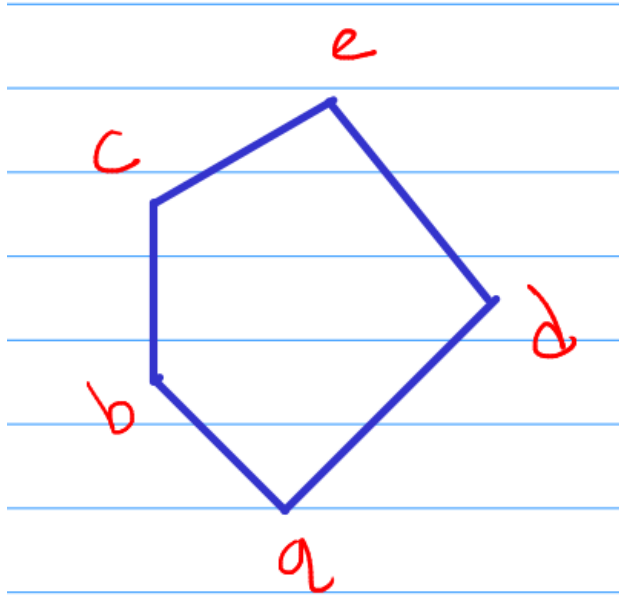
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Chain and Antichain

Chain and Antichain

- Let $(A, \leq)$ be a partially ordered set. A subset of A is called a **chain** if every two elements in the subset are related.
- Let $(A, \leq)$ be a partially ordered set. A subset of A is called an **antichain** if no two elements in the subset are related.

Find Chains and Antichains



Chains :

$\{a, b, c, e\}$

$\{a, b, c\}$

$\{a, d, e\}$

$\{b, c, e\}$

$\{a, b\}$

$\{b, c\}$

$\{c, e\}$

$\{a, d\}$

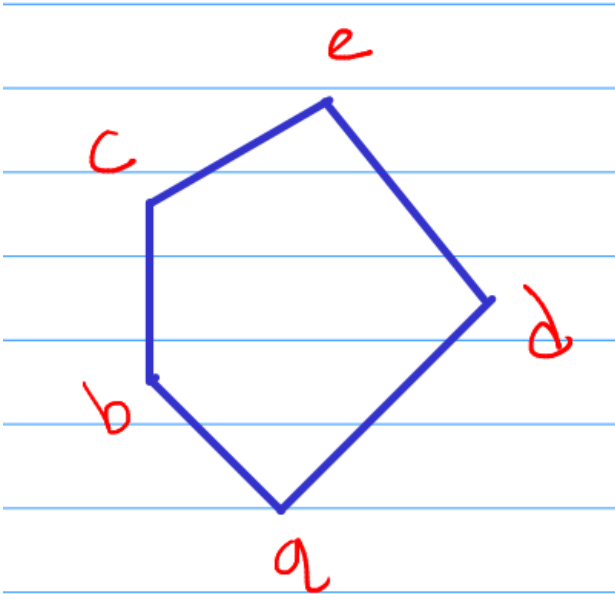
$\{d, e\}$

Find Chains and Antichains

Antichains :

$\{b,d\}$

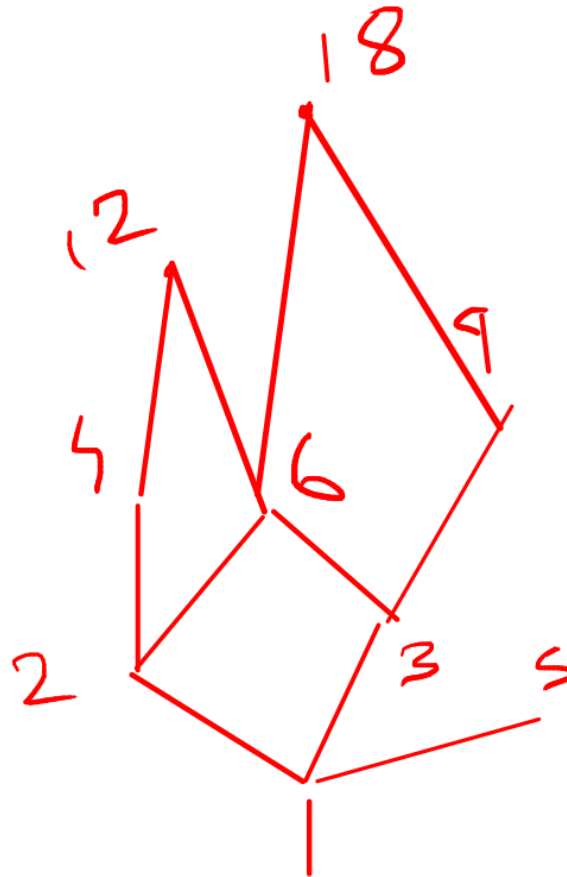
$\{c,d\}$



Example

- $P = \{ 1, 2, 3, 4, 5, 6, 9, 12, 18 \}$, Let R be relation such that (a, b) belongs to R iff a divides b . Find Chains and Antichains.

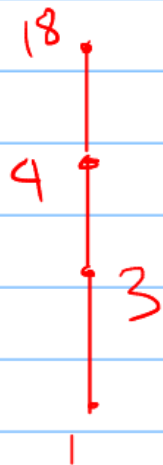
- Chain :
 - $\{1, a\}$ (a belongs to p)
 - $\{1, 2, 6\}$
 - $\{1, 3, 6, 12\}$
 - $\{1, 3, 6, 18\}$
- Antichain
 - $\{3, 5\}$
 - $\{2, 3\}$
 - $\{2, 3, 5\}$
 - $\{3, 4, 5\}$
 - $\{4, 5, 6, 9\}$



Totally Ordered Set

- A partially ordered set $(A, \leq)$ is called totally ordered set if A is a chain.

$$A = \{1, 3, 9, 18\} \quad R = \{(a, b) \mid a \text{ divides } b\}$$



Theorem related to chain - antichain

- **Let $(P, \leq)$ be a partially ordered set. Suppose the length of the longest chains in P is n , then the elements in P can be partitioned into n disjoint antichains.**